Fenchel-Legendre Transform (Fenchel Conjugate)

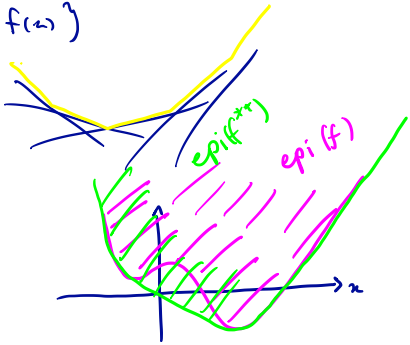
The definition.

$$f : \mathbb{R}^n \to \mathbb{R}$$

$$f^*(y) = \sup_x \{ y^T x - f(x) \}$$

Fenchel conjugate: Important Properties

$$f^*(y) \triangleq \sup_x \{ y^T x - f(x) \}$$



1. f^* is convex.

2. $f(x) + f^*(y) \geq y^T x$

3. $f^{**}(x) \leq f(x) \quad \forall x.$

4. If f is convex & closed, then $f^{**}(x) = f(x)$

5. If f is convex and closed, then the derivatives of f & f^* are inverses of each other:

$$\boxed{y = \nabla f(x) \iff x = \nabla f^*(y)}$$

$$y \in \partial f(x) \iff x \in \partial f^*(y)$$

Fenchel conjugate: An Example

$$f(x) = \frac{1}{2} x^T Q x + q^T x + c, \qquad \underline{Q \succ 0}$$

$$f^*(y) = \sup_x \underbrace{\left\{ y^T x - \frac{1}{2} x^T Q x - q^T x - c \right\}}$$

Take grad, set $= 0$: $\quad y - Qx - q = 0$

$$x^* = Q^{-1}(y - q).$$

$$\Rightarrow f^*(y) = y^T \underbrace{Q^{-1}(y-q)} - \frac{1}{2}(y-q)^T Q^{-1} \cancel{Q Q^{-1}} (y-q) - q^T \underbrace{Q^{-1}(y-q)} - c$$

$$= (y-q)^T Q^{-1}(y-q) - \frac{1}{2} \boxed{} - c$$

$$= \frac{1}{2}(y-q)^T Q^{-1}(y-q) - c.$$

Is f^* convex? YES ✓ Is $Q^{-1} \succ 0$

Gradients are inverses of each other as claimed

- $\nabla f(x) = Qx + q$
- $\nabla f^*(y) = Q^{-1}(y-q)$.

$\nabla f^{-1}(y) = Q^{-1}(y-q)$

Fenchel Conjugate and Duality

$$\min_x : f(x) + g(Ax)$$

$$\min : f(x) + g(y)$$

$$\text{st}: \quad y = Ax$$

$$\min_{x,y} : \max_z : f(x) + g(y) + z^T(Ax - y)$$

if st. duality holds

$$\max_z : \min_{x,y} : (A^T z)^T x + f(x) - z^T y + g(y)$$

$$\max_z : \min_x : -\left[(-A^T z)^T x - f(x)\right] + \min_y : -\left[z^T y - g(y)\right]$$

$$\max_z : -\max_x : (-A^T z)^T x - f(x) - \max_y : z^T y - g(y)$$

$$\boxed{\max_z : -f^*(-A^T z) - g^*(z)}$$

Dual Gradient Ascent

min: $f(x) + g(y)$ // $\max_z: -f^*(-A^T z) - g^*(z)$

st: $y = Ax$

Dual Alg: Gradient Ascent on dual

$$z_{t+1} = z_t + \eta \nabla_z \left(-f^*(-A^T z_t) - g^*(z_t) \right)$$

$$= z_t + \eta \left(A\hat{x} - \hat{y} \right)$$

where: $\hat{x} = \nabla_z f^*(-A^T z_t)$ & $\hat{y} = \nabla g^*(z_t)$

$-A^T z_t = \nabla f(\hat{x})$

$0 = \nabla f(\hat{x}) + A^T z_t$ ////

$\boxed{\hat{x} \in \operatorname{argmin}: f(x) + z_t^T A x}$

$z_t = \nabla g(\hat{y})$

$0 = \nabla g(\hat{y}) - z_t$

$\boxed{\hat{y} \in \operatorname{argmin}: g(y) - z_t^T y}$